

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 6: Lambert Conformal Conic Projection

Syllabus Source: CAE Oxford (pp. 207-248) | Pass Target: $\geq 90\%$ AviationExam

1. THE STANDARD AVIATION CHART ARCHITECTURE

The Lambert Conformal Conic is the primary chart used for aeronautical plotting and mid-latitude IFR airways. It uses a secant cone intersecting the globe at two Standard Parallels (SP1 and SP2):

- Meridians & Parallels: Meridians are straight lines converging at the pole vertex. Parallels are concentric circular arcs centered on the vertex.
- Great Circles on Lambert: GREAT CIRCLES ARE VIRTUALLY STRAIGHT LINES! (Maximum deviation from a straight line over 1,000 NM is negligible).
- Rhumb Lines on Lambert: Rhumb lines are CURVED LINES, concave to the pole of projection.
- Scale Factor (SF): $SF = 1.0$ along the two Standard Parallels. $SF < 1.0$ between the standard parallels (chart is compressed). $SF > 1.0$ outside the standard parallels (chart expands).

2. CONSTANT OF THE CONE & CHART CONVERGENCE

- Constant of the Cone (n): The convergence factor of the cone: $n = \sin(\text{Parallel of Origin}) = \sin(\text{mean latitude of standard parallels})$. E.g. For standard parallels at 30° and 60° , mean lat = 45° , $n = \sin(45^\circ) = 0.7071$.
- Chart Convergence Formula: Chart Convergence (degrees) = Change of Longitude (dlong) $\times$ n = dlong $\times$ $\sin(\text{Parallel of Origin})$

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding Chart Convergence & Great Circle Bearing on Lambert

SCENARIO (Classic ATPL Lambert Chart Drill): A Lambert Conformal Conic chart has standard parallels at 37°N and 65°N , giving a constant of the cone $n = 0.788$:

Waypoint A: $50^\circ\text{N } 020^\circ\text{W}$ - Waypoint B: $50^\circ\text{N } 060^\circ\text{W}$ - A straight line is drawn on the chart from A to B - The measured track angle on the chart at meridian 020°W (Point A) is 260° True

QUESTION: What is the chart convergence between A and B, and what will be the measured track angle at Point B (060°W)?

SOLUTION PASO A PASO (FOR DUMMIES): Step

1: Calculate the Change in Longitude (dlong): - dlong = $060^\circ\text{W} - 020^\circ\text{W} = 40^\circ$. Step

2: Calculate Chart Convergence using the Constant of the Cone (n): - Formula: Chart Convergence = dlong $\times$ n - Chart Convergence = $40^\circ \times 0.788 = 31.52^\circ$ ($\sim 31.5^\circ$)! Step

3: Determine Track Direction Evolution from Point A to Point B: - You are flying WESTBOUND in the Northern Hemisphere. - Meridians converge towards the North Pole at the top of the chart. - As you fly West along a straight line on a Lambert chart, the track angle relative to meridians DECREASES: - Track at Point B = Track at Point A - Chart Convergence

- Track at Point B = $260.0^\circ - 31.5^\circ = 228.5^\circ$ True! FINAL

ANSWER: Chart Convergence = 31.5° . Measured track at Point B = 228.5° True.

[!] EXAM TRAP (AVIATIONEXAM) - Earth Convergence vs Chart Convergence Trap

- Earth Convergence = $d\text{long} \times \sin(\text{lat})$. It varies with every degree of latitude! -
Chart Convergence on Lambert = $d\text{long} \times n$. It is CONSTANT EVERYWHERE on the chart
because 'n' is fixed! - On the parallel where $\sin(\text{lat}) = n$, Earth Convergence equals
Chart Convergence.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On a Lambert Conformal Conic chart, what type of line does a STRAIGHT LINE represent?**

[A] A rhumb line [B] A GREAT CIRCLE (with negligible deviation) [C] A parallel of
latitude [D] An isogonal

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Lambert projection is constructed so that a straight line drawn
between any two points closely approximates a Great Circle track.

Q2: A Lambert chart has a constant of the cone $n = 0.75$. Between longitudes 010°E and 030°E , what is the CHART CONVERGENCE?

[A] 20° [B] 15° ($d\text{long} = 20^\circ \times 0.75 = 15^\circ$) [C] 7.5° [D] 30°

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Chart Convergence = $d\text{long} \times n = (30^\circ - 10^\circ) \times 0.75 = 20^\circ \times 0.75 = 15^\circ$.

Q3: Where on a Lambert Conformal Conic chart is the Scale Factor (SF) LESS THAN 1.0?

[A] North of the northern standard parallel [B] BETWEEN THE TWO STANDARD PARALLELS [C]
Outside both standard parallels [D] Exactly along the standard parallels

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Because the cone cuts inside the sphere between the two standard
parallels, the map features are slightly compressed ($SF < 1.0$). Along the standard
parallels, $SF = 1.0$.